

August Herron

augustherron.com | ryderherron@gmail.com | 512-717-1253 | Austin, TX

PURPOSE

Driven computer science and physics undergraduate looking to gain professional experience in the fields of math, physics, and computer science.

EDUCATION

Trinity University | 3.77 GPA

B.S. Computer Science (3.97 GPA), B.S. Applied Physics (3.93 GPA), Minor in Math

San Antonio, TX

August 2022 – May 2026

PROJECTS

Chaotic Pendulum Simulation Web App | *TypeScript, HTML/CSS, PixiJS, Plotly.js*

- Built a double pendulum simulation web app showcasing non-linear systems and chaotic dynamics.
- Used HTML/CSS and TypeScript along with libraries such as PixiJS and Plotly.js to make an interactive visual simulation, displaying the pendulum itself as well as phase portraits.
- Showed the derivation of the equations of motion using Lagrangian Mechanics and used Runge-Kutta (RK4) integration to numerically solve the governing set of coupled non-linear ordinary differential equations.
- Built using Webpack and hosted on my personal website using GitHub pages.

Highpass/Lowpass Filter Audio Plugin | *C++, JUCE*

- Created a custom highpass and lowpass filter audio plugin that can be used in digital audio workstations, such as Ableton Live, for use in audio processing and music production.
- Built using the JUCE C++ audio plugin development framework.

Partial Compiler | *C*

- Implemented components of a source to source compiler for the assembly language ILOC.
- Implements a lexer, parser, register allocator, and instruction scheduler.
- Created for my course on compiler construction using the C language.

Linear Regression ML Model | *Python, pandas, Seaborn, Matplotlib*

- Built a linear regression ML model from scratch to predict insurance cost using BMI and smoker status.
- Used the pandas Python library to read, sort, and manage data.
- Used the Seaborn and Matplotlib Python libraries to visualize results.

EXPERIENCE

Teaching Assistant

Trinity University

August 2024 – Present

San Antonio, TX

- Teaching assistant and tutor for Calculus III at Trinity's Quantitative Reasoning and Skills center.

Lifeguard

City of Austin

June 2023 – August 2023

Austin, TX

Engineering Intern

AECOM

June 2021 – July 2021

Austin, TX

- Civil engineering internship about the Orange Line light rail system being designed in Austin.
- Used CAD software to design a light rail station with 3 other interns.
- Presented the proposed light rail station to the engineers and executives at AECOM and Capital Metro in Austin.

SKILLS

Programming languages: C++, C, Python, Java, HTML/CSS, JavaScript/TypeScript

Libraries/Frameworks: JUCE, pandas, NumPy, Seaborn, Matplotlib, Plotly, PixiJS

Miscellaneous: Git, Linux, DSP, Latex, CAD, Digital Electronics, Mathematics and Physics, MS Excel/Word/Powerpoint

RELEVANT COURSEWORK

Computer Science: Data Structures and Algorithms, Compiler Construction, Operating Systems

Mathematics: Linear Algebra, Differential Equations, Calculus I, II and, III, Abstract Math, Numerical Analysis

Other: Digital Electronics, Electricity & Magnetism, Mechanics, Modern Physics